2. Creating Your First R Script

In this section, you will create your first R script.

What is an R Script?

An **R script** is a file where you write R code.

R script files end with:

```
.R
```

For example:

```
first_script.R
```

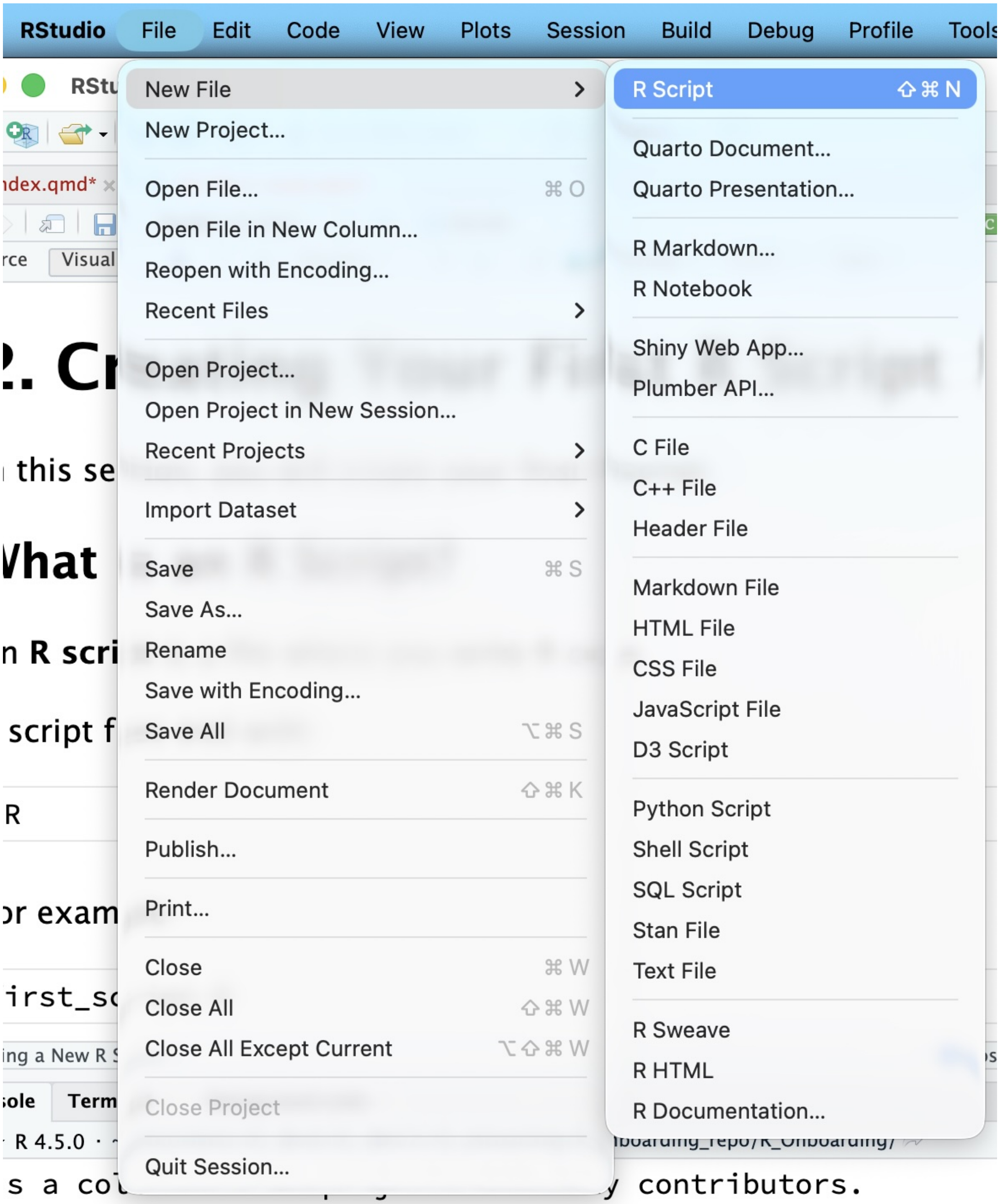
An R script is useful because it saves your code. You can run the same code again later without retyping everything.

> Tip
>
> Use an R script when you want to save your work.
> Use the Console when you want to quickly test one command.

Creating a New R Script

In RStudio:

1. Click **File**.
2. Click **New File**.
3. Click **R Script**.



A blank script should open in the Source Pane.

Basic Arithmetic

Before we work with data, let's try R as a calculator.

Type the following into your script:

```r
2 + 3
```

```
[1] 5
```

```r
10 - 4
```

```
[1] 6
```

```r
5 * 6
```

```
[1] 30
```

```r
20 / 4
```

```
[1] 5
```

These symbols mean:

Symbol	Meaning
`+`	Add
`-`	Subtract
`*`	Multiply
`/`	Divide

> **Try It Yourself**
>
> Change the numbers and run the code again.
> For example, try:
>
> ```r
> 100 / 5
> 12 * 3
> ```

Creating Variables

A **variable** is a name that stores a value.

For example:

```r
age <- 18
age
```

```
[1] 18
```

The symbol `<-` means "store this value."

So this line:

```r
age <- 18
```

means:

> Store the number 18 in a variable called `age`.

Why Use Variables?

Variables help us reuse information.

For example, instead of typing the same number many times, we can give it a name.

```r
quiz_score <- 92
homework_score <- 88

quiz_score
```

```
[1] 92
```

```r
homework_score
```

```
[1] 88
```

Now we can use those variables in a calculation:

```r
final_score <- (quiz_score + homework_score) / 2
final_score
```

```
[1] 90
```

This code calculates the average of the quiz score and homework score.

Printing Output

Sometimes you want to clearly print something.

You can use `print()`:

```r
print(final_score)
```

```
[1] 90
```

For simple work, typing the object name also shows the value:

```r
final_score
```

```
[1] 90
```

Both are okay.

> **Tip**
>
> In R, object names should not contain spaces.
> Use this:
>
> ```r
> quiz_score <- 92
> ```
>
> Not this:
>
> ```r
> quiz score <- 92
> ```

Adding Comments

A **comment** is a note to yourself or your reader.

In R, comments start with **#**.

R ignores comments when running code.

```
# This stores my quiz score
quiz_score <- 92

# This prints the quiz score
quiz_score
```

```
[1] 92
```

Comments are useful because they help explain what your code is doing.

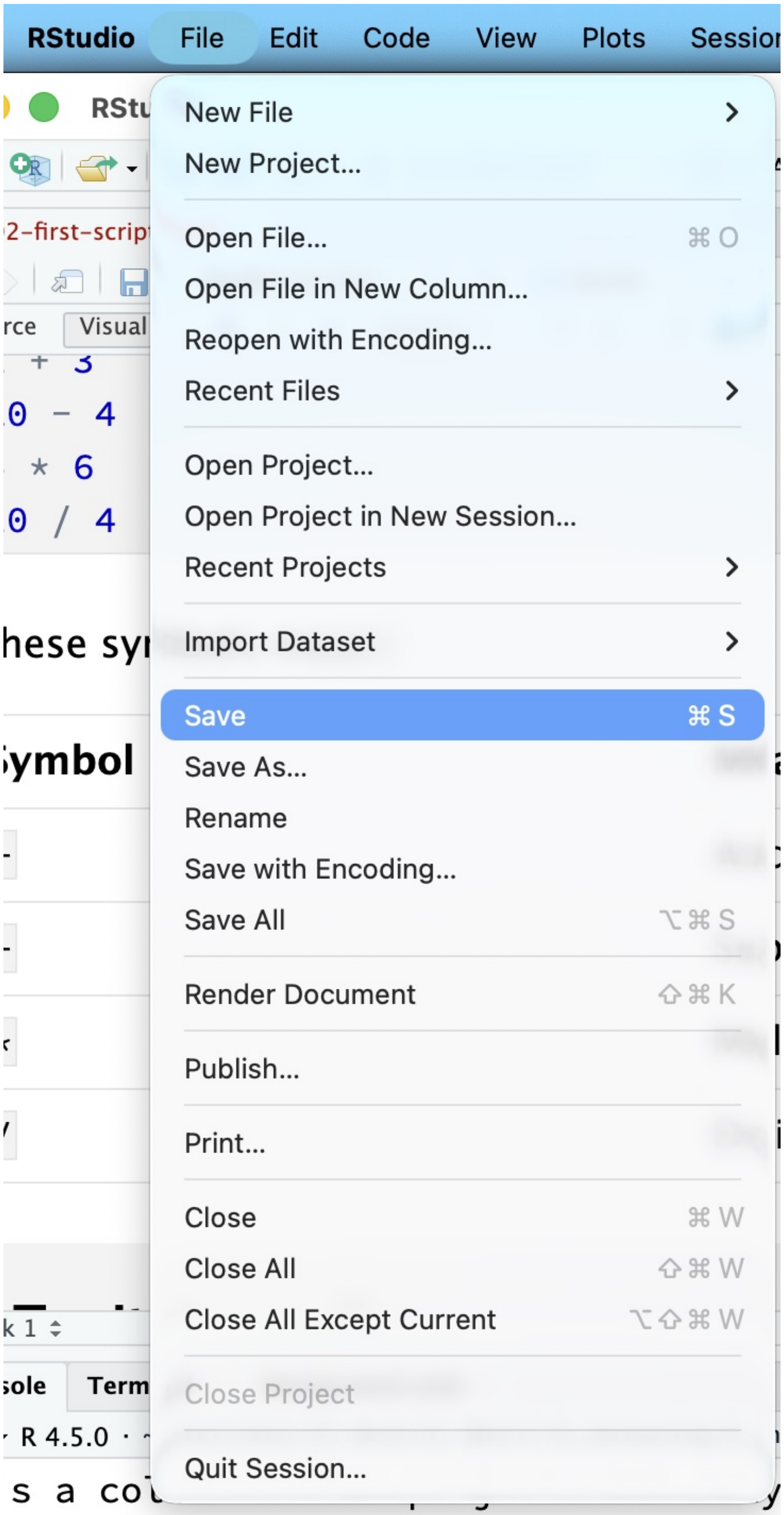
Saving Your Script

To save your script:

1. Click **File**.
2. Click **Save**.
3. Name your file:

```
first_script.R
```

4. Save it in your project folder.


RStudio
File
Edit
Code
View
Plots
New File
New Project...
Open File... ⌘ O
Open File in New Column...
Reopen with Encoding...
Recent Files
Open Project...
Open Project in New Session...
Recent Projects
Import Dataset
Save ⌘ S
Save As...
Rename
Save with Encoding...
Save All ⌥ ⌘ S
Render Document ⇧ ⌘ K
Publish...
Print...
Close ⌘ W
Close All ⇧ ⌘ W
Close All Except Current ⌥ ⇧ ⌘ W
Close Project
Quit Session...

Show the Save button and file name `first_script.R`.

> ⚠ Common Mistake
>
> Do not save your script with a name like:
>
> ```
> Untitled1.R
> ```
>
> Use a clear name, such as:
>
> ```
> first_script.R
> ```

Practice Exercise

> ℹ Practice
>
> Create a new R script and write code that:
>
> 1. Stores your favorite number in a variable.
> 2. Stores your age in another variable.
> 3. Adds the two numbers together.
> 4. Prints the result.
>
> Example:
>
> ```
> favorite_number <- 7
> age <- 18
> result <- favorite_number + age
> print(result)
> ```

Check Your Understanding

Answer these questions in your own words:

1. What is an R script?
2. Why is it useful to save code in a script?
3. What does `<-` do?
4. What is the difference between these two lines?

```
score <- 90
print(score)
```

Summary

In this section, you learned how to:

- Create a new R script.
- Use R for basic arithmetic.
- Store values in variables.
- Print results.
- Add comments.
- Save your script.

You have now written your first R code!